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# Direct optical thermometry in semiconductors

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N. Roubinowitz,[1] T. Vezin,[2] M. Legrand,[2] S. Der,[3] B. Yu Jeco-Espaldon,[2,4] S. Rani,[1] M. Giteau,[5] Y. Okada,[2] J. F. Guillemoles,[1] S. Houver,[3] and D. Suchet[1,a)]

## AFFILIATIONS

[1]Institut Photovoltaique d'Ile de France, UMR-IPVF 9006, CNRS, Ecole Polytechnique IPP, ENSCP PSL, Palaiseau, France
[2]Research Center for Advanced Science and Technology, The University of Tokyo, Meguro-ku, Tokyo 153-8904, Japan
[3]Université Paris Cité, CNRS, Laboratoire Matériaux et Phénomènes Quantiques, Paris, France
[4]Center for Research and Innovations in Science, Mathematics, and Education, West Visayas State University, Iloilo City, Iloilo 5000, Philippines
[5]Laboratoire PROcédés, Matériaux et Énergie Solaire (PROMES), CNRS, Odeillo, France

[a)]**Author to whom correspondence should be addressed:** daniel.suchet@polytechnique.edu

## ABSTRACT

This pedagogical review offers an easy access to three direct optical methods (photoluminescence, absorption, and Raman spectroscopies) to probe temperatures in out-of-equilibrium semiconductor devices. For each technique, starting from basic considerations, the operating principles and usual assumptions are clarified; the strategy to acquire and analyze data is explained, and the most critical points of attention are underlined with examples. Decision trees and a comparison table are provided to help identify the best suited approach to specific characterization needs.

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## I. INTRODUCTION

Semiconductor devices can be driven out of equilibrium by a variety of external excitations—most often optical illumination or electric bias. As the imparted energy is redistributed among electrons, holes, and phonons through rapid interactions, each ensemble relaxes quickly toward quasi-thermal states, which can be described with a temperature and a chemical potential. Ultimately, interactions between these subsystems will result in their complete thermalization toward shared thermodynamic properties. In some cases however, the coupling between these subsystems is weak enough so that they may remain out of equilibrium with each other. Typically, charge carriers can acquire energies corresponding to an effective temperature well above that of the surrounding lattice ("hot carriers"), while distinct phonon populations may themselves remain out of equilibrium with both carriers and the environment.

These non-equilibrium situations occur in a range of semiconductor devices and energy conversion systems. For instance, in high-field transistors, a strong electric field may generate hot carriers, which can be detrimental to transport characteristics and device reliability.[1–3] Conversely, in hot-carrier solar cells, reduced carrier cooling has been considered to engineer an electronic population hotter than the lattice, thereby enhancing the power conversion efficiency.[4] Such hot carriers have also been discussed in the context of photocatalysis.[5] Similarly, thermionic cooling relies on the engineering of a cold electronic population, which extracts heat from the lattice through electron–phonon interactions.[6] These temperature disparities critically affect transport, recombination, and energy conversion processes and are central to the operation and optimization of such devices.

To study and understand the thermalization[7] properties, and to consider specific applications, it is crucial to have simple and reliable techniques to estimate and discriminate the temperatures of the different constituents of these systems.

Optical techniques have proven particularly well suited for thermometry.[8–10] They can monitor a broad range of temperatures (from cryogenic temperatures to thousands of Kelvins) with a high degree of accuracy (down to the mK for some). Experimentally, they provide a contactless approach that can be readily implemented. This makes them especially valuable at early stages of device development, where optimized interfaces or fully engineered architecture may not yet be available. By relying on the interaction between light and matter, optical thermometry offers a flexible and versatile means to


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probe temperature, with the potential for high spatial and temporal resolution.[11]

Optical thermometry encompasses a variety of techniques. *Indirect* approaches rely on the temperature-dependent response of reference probes, which are assumed to remain in thermal equilibrium with the system under scrutiny. For instance, quantum dots can be embedded into a material during fabrication and used as local thermometric markers: once their luminescence has been calibrated as a function of temperature, the emitted radiation provides an *in situ* measure of the local temperature.[12,13] A similar principle underlies fluorescence thermometry in optical fibers,[14] where the well-characterized temperature dependence of the emission from rare-earth-doped fibers enables their use as versatile and distributed temperature sensors. Another example is photothermal deflection spectroscopy,[15] in which the sample is immersed in a medium with a temperature-dependent refractive index; the deflection of a probe beam near the sample surface reveals local temperature gradients, which can then be integrated to infer the surface temperature.

In this pedagogical review, we focus on three optical techniques where the temperature is measured *directly*, through the spectral response of the system of interest. All three techniques can either be performed in transient mode (TR), to follow thermalization dynamics, or under continuous-wave illumination (CW), to assess a steady-state temperature distribution.

In the *transient absorption spectroscopy* (TAS) process, sketched in Fig. 1(a), an incident monochromatic beam (green arrow) photoexcites electrons from the valence to the conduction band. A second beam (red arrow) is sent across the sample and will be absorbed depending on the occupation factors of the addressed transition. Carriers' distribution is, thus, probed by measuring the absorptance of the sample—i.e., the ratio between the incident and absorbed powers.

In the *photoluminescence* (PL) process, sketched in Fig. 1(b), an incident monochromatic beam (green arrow) photoexcites electrons from the valence to the conduction band. Among the processes that allow the relaxation of the excited population, radiative recombination expels the energy of the electron–hole pair in photon emission (red arrow), which can be measured spectrally.

In the *Raman spectroscopy* process, sketched in Fig. 1(c), an incident monochromatic beam (orange arrow) of frequency and direction ($\omega_L$, $k_L$) interacts with a material, resulting in scattered light at ($\omega_S$, $k_S$), with $\omega_S \neq \omega_L$, referred as inelastic scattering. This process induces a transition of an intrinsic material excitation, typically a vibrational excitation. In the case of excitation creation or destruction, the process is called Stokes or anti-Stokes Raman scattering, respectively, leading to a corresponding scattered light with either $\omega_S < \omega_L$ or $\omega_S > \omega_L$. These processes are represented with red and yellow arrows, respectively.

While these techniques are all routinely employed, they are often developed and used within distinct scientific communities, with limited interaction between them. Yet, they are highly complementary and could, when combined, provide a more comprehensive picture of temperature distributions in complex systems, since they give access to different subsystem temperatures, like carriers and phonons. Moreover, a clear understanding of their respective principles and limitations is essential for meaningfully comparing results obtained through different approaches.

The overarching aim of this tutorial review is, therefore, to provide a unified framework to navigate these methods, building on a solid understanding of the underlying principles. We emphasize that the scope of this work is focused on temperature estimations, not on the microscopic cooling mechanism resulting in these temperatures.[10] For each technique, we first introduce the working principles to clarify which subsystem is effectively being probed. We then discuss the most common sources of error that may bias temperature measurements, along with methodological strategies to mitigate them. This discussion shines light on standard data analysis procedures and their underlying assumptions, which are often left implicit and may be applied beyond their range of validity. Finally, comparing these techniques may help identify which one may be best suited for specific applications.

We emphasize that the methods discussed in this review *assume* that the ensemble under scrutiny follows a thermal distribution (i.e., Fermi–Dirac or Bose–Einstein distribution) and aim at estimating its temperature. This assumption may be invalid if the system follows a non-thermal distribution (e.g., at very short times in time-resolved methods or under very strong forcing in steady-state methods), where the very notion of temperature is not a valid thermodynamic quantity. The validity of this key assumption must be carefully verified for the device and experimental conditions under consideration. A possible sanity check is to evaluate whether a thermal distribution is indeed able to account correctly for the experimental observation or not.[16]

## II. ABSORPTION SPECTROSCOPY

Absorption spectroscopy consists in probing the absorptance $A$ of a given material. In the context of thermometry, this measurement is most often conducted under an optical excitation, which generates charge carriers in the bands. The photo-induced carrier distribution modifies the absorptance of the system, which provides insights into carrier distributions.[17] Absorption spectroscopy can be conducted in similar ways in the steady-state and time-resolved regimes, although the vast majority of the literature focuses on probing carrier dynamics with time-resolved measurements.

### A. Working principle

In the general case, the absorption coefficient can be computed from matrix elements and occupation functions of the different bands through the Fermi golden rule,[18]

$$\alpha(E) \propto \iint d^3k_h d^3k_e |M_{eh}|^2 (1 - f_h(k_h) - f_e(k_e)) \delta(E - E_h(k_h) - E_e(k_e)), \tag{1}$$

where $\alpha$ is the absorption coefficient of the material (in $m^{-1}$), $E$ is the energy of the optical transition under consideration, $k_h$ (respectively, $k_e$) is the wavevector of the hole (respectively, electron) state of the transition at energy $E_h$ (respectively, $E_e$), $M_{eh}$ is the matrix element of the transition, $f_h$ (respectively, $f_e$) is the occupation function of holes (respectively, electrons) in the valence (respectively, conduction) band, and $\delta$ is the usual Dirac function. The energy convention used in this review is such that

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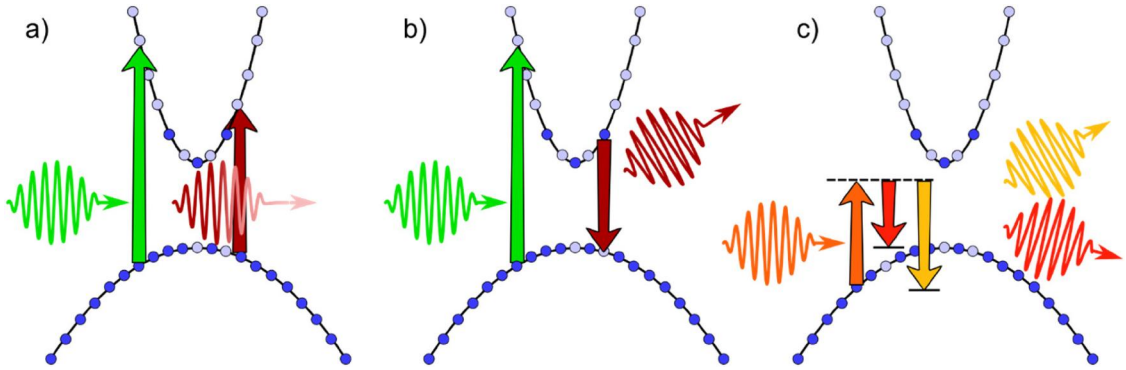



**FIG. 1.** Energy diagram presenting the three thermometry techniques discussed in this work, (a) the transient absorption spectroscopy, where the pump is followed by a second broadband excitation, (b) the photoluminescence, where the pump induces emission of light at lower photon energy, and (c) the Raman spectroscopy, where the pump leads to low-energy modes excitations via Stokes and anti-Stokes processes.

both $E_e$ and $E_h$ are positive and referenced to the mid-gap energy, as in Ref. 19.

Upon light absorption, two kinds of effects occur simultaneously. First, photoexcited carrier distributions reduce the absorption coefficient by increasing the occupation functions of electrons and holes.[17] This effect is called either "occupation effect," "band filling," or "state filling." It is the physical phenomenon yielding direct information on the distributions. Second, at sufficiently high carrier densities, additional "many-body effects" will manifest,[17,20] such as bandgap renormalization (reduction of the bandgap due to high carrier densities in the bands),[21] linewidth broadening (broadening of the absorption edges, partially explained by the interaction with LO phonons),[22,23] and exciton screening (screening of the emission of exciton states[20,24] by free carriers).

The effects of band filling and bandgap renormalization are illustrated in Fig. 2. The density and temperature of excited carriers modify the absorption coefficient as compared to the equilibrium situation [Figs. 2(a) and 2(b)]. In turn, this change impacts the absorptance and transmittance of the system [Figs. 2(c) and 2(d)], which can be monitored to evaluate the carriers' distribution properties.

Two salient features can be highlighted. First, note that band filling is mainly visible for energies around the quasi-Fermi level splitting (QFLS) $\Delta\mu$, resulting in a decrease in α between 1.1 and 1.3 eV in this example. This decrease becomes more pronounced as $\Delta\mu$ increases and is accompanied by a slight blue shift of the spectral extrema of $\Delta\alpha$ and $\Delta T$. When injection is large enough such that $\Delta\mu$ becomes larger than the bandgap, population inversion occurs and the absorption coefficient becomes negative below $\Delta\mu$ (corresponding to an optical amplification), as shown in Fig. 2(a) with the dashed curves. In any case, the effect of band filling is noticeable up to several times $k_B T_c$ above the quasi-Fermi level splitting $\Delta\mu$, which makes it possible to detect carrier temperature. Second, bandgap renormalization leads to a shift of the density of states toward lower energies. This has two distinct effects on the spectrum as it creates new states for absorption at low energy, but also increases the absorption coefficient at high energy due to the shifted density of states. This latter effect is sometimes called "photo-induced absorption" (PIA) and will be described in greater detail in Sec. II D 5.

Assuming that electrons and holes follow thermal distributions, the temperature of carriers is a parameter of Eq. (1) through the band filling term $1-f_h-f_e$. This can also be seen in Fig. 2, where plain and dashed blue lines represent different carrier temperatures (respectively, $T_c = 300$ and 600 K). Lattice temperature might also be measured from this formula as lattice temperature changes the bandgap (see Sec. III C 2), through the matrix elements $M_{eh}$.[25] Note, however, that we did not identify any report of a direct measurement of lattice temperature through absorption spectroscopy. This can possibly be explained by the fact that a change of lattice temperature may have a similar effect on the absorption coefficient as bandgap renormalization, making the discrimination between the two effects challenging. This section will, therefore, focus on the measurement of carrier temperature from now on.

## B. Experimental setup

Experimentally, the absorption coefficient cannot be accessed directly. It is indeed more straightforward to measure integrated quantities such as the absorptance $A$, which can be deduced from a measurement of the transmittance $T$ and the reflectance $R$,

$$A = 1 - T - R. \tag{2}$$

Additional hypotheses are then necessary to link the variations of $A$ to those of $\alpha$ (see Secs. II C and II D for details).

### 1. Transient absorption spectroscopy

Nowadays, this technique is commonly referred to as "transient absorption spectroscopy" (TAS). However, it was first known as "flash photolysis,"[26] and several alternative names still exist to designate this technique, such as "saturation optical absorption

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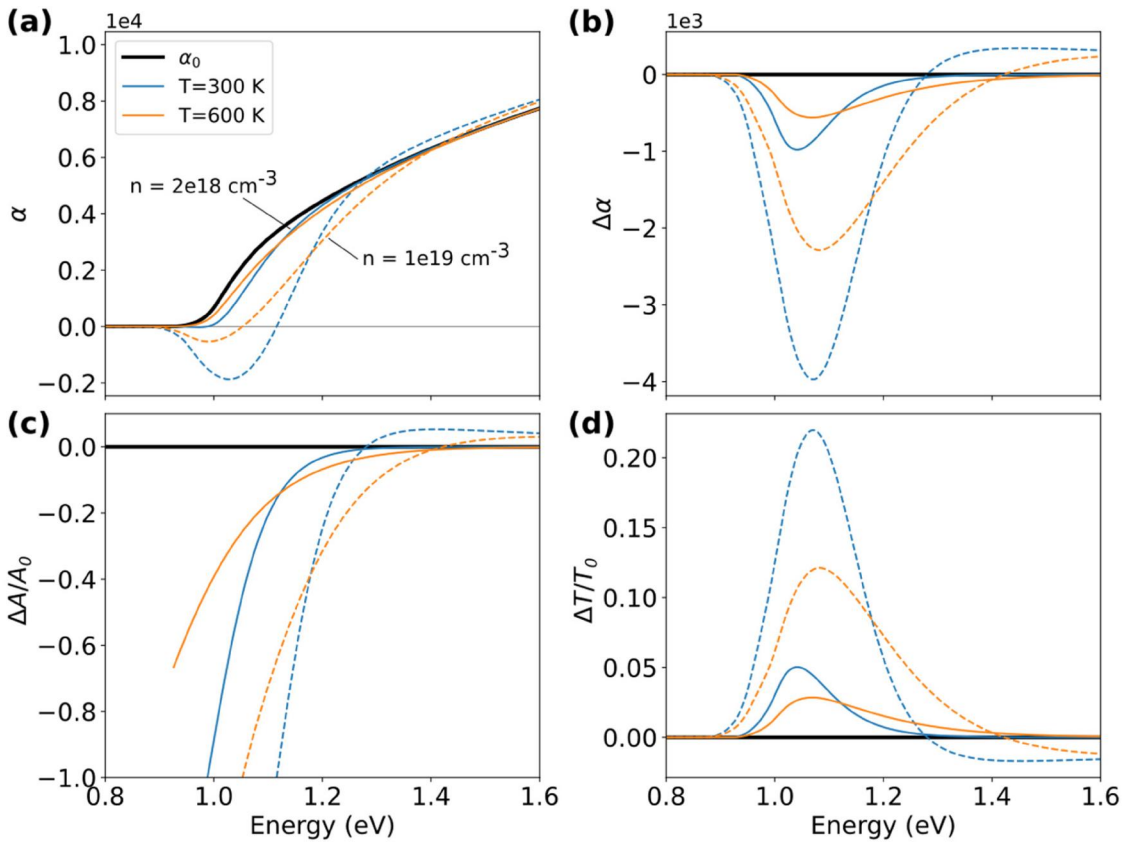



**FIG. 2.** Numerical computation of the photoresponse of a direct gap semiconductor with $E_g = 1$ eV under light bias. Black line corresponds to the reference case in absence of pump. Blue curves are computed with a carrier temperature of $T_c = 300$ K while orange curves consider $T_c = 600$ K. Solid (respectively, dashed) curve are obtained with carrier density $n = 2 \times 10^{18}$ cm$^{-3}$ (respectively, $n = 10^{19}$ cm$^{-3}$), representing mild (respectively, strong) band filling effect. (a) Absorption coefficient ; (b) Variation of the absorption coefficient as compared with unbiased case $\Delta\alpha = \alpha - \alpha_0$ ; (c) Normalized variation of the absorptance $A = 1 - \exp(-\alpha d)$ ; (d) Normalized variation of the transmittance $T = \exp(-\alpha d)$ (assuming negligible change in reflectance).

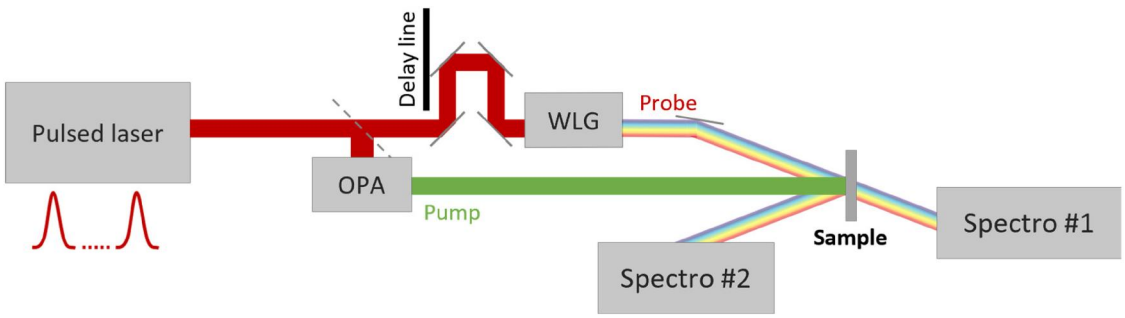



**FIG. 3.** Principle of transient absorption spectroscopy. A pulsed laser is used to generate both the pump and probe beams. The pump wavelength is selected by an optical parametric amplifier (OPA), while the probe beam is produced by a white light generator (WLG). In this example, both the reflectance and transmittance are acquired to rigorously measure the absorptance.

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pump-and-probe technique"[27] or "dynamic absorption spectroscopy."[28]

In any case, experimental setups for measuring TAS use a pulsed laser as a pump to excite the sample and then use another pulse of broadband light to probe the material with a given time delay. The transmitted and/or reflected beams are collected by one or several spectrometers to analyze their spectral dependency. The temporal resolution of typical commercial setups is around 100 fs.[29,30]

Figure 3 depicts a typical TAS setup where one laser is split to provide both the pump and probe beams. This approach is simpler and more commonly found than setups based on the synchronization of two different lasers. To measure changes induced by the pump, the probe should not affect the signal significantly and must, therefore, have a much lower intensity than the pump. Furthermore, the probed area should be smaller than the pumped area to only probe the photoexcited part of the sample.

Depending on the experiment, the transmitted and/or reflected probe pulse can be measured.[31] These signals are analyzed by comparing the magnitude of the probe pulse transmitted/reflected with $(T, R)$ and without $(T_0, R_0)$ the pump beam. Hence, the quantities of interest are $\Delta T = T - T_0$ and $\Delta R = R - R_0$. In principle, both measurements are necessary for rigorous measurement of the absorptance change, as $\Delta A = -(\Delta T + \Delta R)$.[32] However, many authors measure only $\Delta T$ by assuming negligible or constant reflectance. The validity of this assumption will be further discussed in Sec. II D 4.

It is nowadays most common to use broadband probe beams in combination with spectrometers to measure carrier temperature directly from the shape of the transient absorption spectra. However, several historical studies reported indirect measurement of carrier temperature from monochromatic measurements.[33,34]

### 2. Steady-state pump-probe absorption spectroscopy

An alternative version of the setup shown in Fig. 3 could in principle be implemented with a continuous-wave laser in place of the pulsed source. To date, however, only one study has explicitly extracted carrier temperature from the steady-state, pump-induced modification of the near-edge absorption spectrum of a semiconductor.[35] In that work, a tuned broadband dye laser simultaneously provided the pump—just above the bandgap—and the probe at lower photon energies.

## C. Carrier temperature measurement—Ideal case

In this section, we assume that the change in absorptance $\Delta A$ of a sample under a pump beam has been acquired with a spectrometer, as described in Sec. II B. We will describe the main hypothesis, allowing us to deduce the temperature of carriers from this measurement.

First, we consider that $A(E) \propto \alpha(E)$. This approximation is often invoked the case in systems with low absorption, as the Beer–Lambert law leads to

$$A(E) = (1 - R)(1 - \exp(-\alpha(E)d)) \simeq (1 - R)\alpha(E)d, \tag{3}$$

where $d$ is the optical depth of the absorber. Note, however, that this expression neglects multiple reflections and optical interferences, which may be significant in thin films. An inaccurate optical model may result eventually in an uncontrolled error in the temperature estimation.

Second, we neglect many-body effects, such that matrix elements from Eq. (1) and local density of states can be considered independent of carrier distributions.

Finally, we consider the effective mass approximation, so that the dispersion relations of both electrons and holes are treated as parabolas. Accordingly, individual electron and hole energies can be expressed as a function of the corresponding optical transition energy $E$ only:[19]

$$\begin{cases} E_e = \dfrac{E_g}{2} + (1 - \xi)(E - E_g), \\ E_h = \dfrac{E_g}{2} + \xi(E - E_g), \end{cases} \tag{4}$$

where we use the effective mass mismatch parameter $\xi = m_e/(m_e + m_h)$. In that case, the term $(1 - f_e - f_h)$ can be taken out of the integral in Eq. (1) and the absorption coefficient, therefore, is written as[17]

$$\alpha(E) = \alpha_0(E)[1 - f_h(E_h) - f_e(E_e)], \tag{5}$$

where $\alpha_0$ corresponds to the intrinsic absorption coefficient without pump excitation. Note that the effective mass approximation, while being customary for TAS analysis, may not apply to the material under scrutiny—in which case, a different relationship between the absorption coefficient and the carriers' temperature should be considered.

Combining those assumptions, and assuming that the reflectance is unaffected by the pump (i.e., $R = R_0$),

$$\frac{\Delta A}{A_0} = \frac{\Delta \alpha}{\alpha_0} = -(f_e(E_e) + f_h(E_h)). \tag{6}$$

Therefore, the TAS signal can be simply linked to the sum of the occupation function of electrons and holes. Note that many authors prefer to study variations of the transmittance. With same hypotheses as above (the assumption that reflectance is unchanged will be discussed below), transmittance changes as

$$\frac{\Delta T}{T_0} = -\Delta \alpha d = \alpha_0 d(f_e(E_e) + f_h(E_h)). \tag{7}$$

For thermal distributions, the occupation factors follow the Fermi–Dirac distribution, and the previous quantities can be expressed from Eq. (4). This development can be found in the earlier days of TAS (see, e.g., Refs. 33, 36, and 37 on GaAs). More recent studies assume that one of the carrier types is much lighter than the other. This is typically the case in most III–V materials, where $m_e \ll m_h$ (i.e., $\xi \ll 1$), such that the dominant energy

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dependence of the TAS signal is

$$\frac{\Delta A}{A_0}(E) \propto -f_e(E_e) = -\left[\exp\left(\frac{E-\mu_e-\frac{E_g}{2}}{k_B T_e}\right)+1\right]^{-1} \times \propto -\exp\left(-\frac{E}{k_B T_e}\right). \quad (8)$$

This expression can be a good approximation of the TAS signal in III–V materials with strong mass mismatch, as in Ref. 38. However, it has also been heedlessly used in the context of perovskite materials,[29,32,39–43] where $m_e \simeq m_h$ (i.e., $\xi \simeq 1/2$). In this case, an improper use of Eq. (8) leads to an overestimation of the temperature by as much as a factor 2, as the correct expression should be

$$\frac{\Delta A}{A_0}(E) = -(f_e(E_e)+f_h(E_h)) \propto -\exp\left(-\frac{E}{2k_B T_c}\right), \quad (9)$$

where $T_c = T_e = T_h$ is the carrier temperature. Note that this overestimation does not affect the estimation of the carrier *cooling time*.

## D. Limits, biases, and possible corrections

In this section, we discuss the limitations of the simple method presented in Sec. II C and present several attempts to go beyond.

### 1. Sample requirements

In order to measure the change in transmittance, the sample (both the absorber and the substrate) needs to be transparent enough in the wavelength range of investigation to obtain sufficient signal quality. This is all the more important considering the need for a suitable fitting interval as detailed in Sec. II D 2. Samples may be placed on a substrate that is transparent for the probe beam energies (e.g., glass, quartz, high-gap semiconductor, etc.), or transferred onto such substrates, which may require additional fabrication steps and the use of resin. It is, thus, recommended to perform a control experiment on the bare substrate (or substrate and resin) to identify their potential contribution to the measured signal. In a more anecdotal way, it is possible to fabricate free standing films, for instance, by locally etching the substrate.[44]

The sample must also allow efficient heat dissipation to avoid heating up the sample's lattice to temperatures higher than ambient, complicating the temperature interpretation. Furthermore, there is a risk of transient lattice heating affecting the TAS spectra.[45] It is, therefore, of importance to favor good thermal contact between the sample and the substrate, so that the substrate acts as a thermal sink. A rotating sample mount can be used to mitigate local heat accumulation.[46]

### 2. Fitting interval

The first drawback of a simple exponential fitting of TAS signal is that it is very sensitive to the choice of the fitting interval.[47]

To be valid, these fittings must occur over an energy window where the band filling dominates. However, at large energies ($E-\Delta\mu \gg k_B T_c$), band filling becomes negligible and other effects such as photo-induced absorption dominate (see Secs. II D 3 and II D 4). Similarly, at low energies ($E \simeq E_g$) and high fluences, many-body effects such as bandgap renormalization and exciton screening dominate (see Sec. II D 5).[39] The fitting window is, therefore, strongly constrained.

Several authors pointed out the difficulty of defining rules to identify a valid fitting interval over a large collection of experimental spectra. For instance, Yang *et al.*[48] studied the impact of the width of the fitting interval and concluded that an interval of at least 0.2 eV was necessary for reliable temperature measurements. Lim *et al.*[49] refined the analysis by studying different fitting schemes and showed that different choices in fitting interval can change the estimated carrier temperature by a factor 2 in some cases. Corresponding uncertainties range from several tens[48] to several hundreds of kelvin.[49]

Overall, this difficulty in defining a valid fitting window makes exponential fitting deceptively simple and quite unreliable. More refined methods are needed in order to extract reliable temperatures, as will be shown in Sec. II D 5.

### 3. High-energy photo-induced absorption

Frequently, transient transmittance spectra are negative at high energy, an effect often called "photo-induced absorption" (PIA). This effect makes it impossible to apply simple exponential fittings, as the exponential model does not change sign at high energy.

Most authors treat this PIA as a negative constant in the standard exponential fit,[29,40,41,47,50] without presuming on its physical meaning. This allows them to extend the fitting range toward higher energies, thus reducing the sensitivity of the fit to the fitting interval.

On the other hand, several authors suggested that this PIA might stem from bandgap renormalization.[48,51] Indeed, bandgap renormalization results in a shift of the density of states (thus of the absorption coefficient) toward lower energies. As the absorption coefficient is typically a growing function of the energy, this leads to an increase of the absorption coefficient for energies where the band filling is sufficiently weak, that is, at high energy. This is exemplified in Figs. 2(c) and 2(d) (dashed curves). With an ideal bulk absorption coefficient, the first-order correction can be expressed as

$$\frac{\Delta A}{A_0} = \frac{\Delta\alpha}{\alpha_0} = \sqrt{1+\frac{\Delta E_g}{E-E_g}}(1-f_e-f_h)-1 \simeq -(f_e+f_h)+\frac{1}{2}\frac{\Delta E_g}{E-E_g}. \quad (10)$$

In practice, several authors have used instead a phenomenological correction scaling as $E^{-1/2}$ to account for this PIA.[39,48,51] It is also possible to treat the effect of bandgap renormalization through a more rigorous full fit approach, as will be shown in Sec. II D 5.

Note, however, that bandgap renormalization is not the only effect that can explain photo-induced absorption. This feature can also be derived from photo-induced change in the reflectance signal.

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#### 4. Changes in refractive index and reflectance

To measure properly the changes in absorptance, it is necessary to monitor both the changes in transmittance and reflectance of the sample under pump excitation. However, most authors measure only variations of the transmittance out of simplicity and, thus, neglect changes in the reflectance (most of the time implicitly). In this section, we show that this approximation does not hold in some cases and may lead to an incorrect measurement of the temperature.

In fact, the absorption coefficient $\alpha$ and the refractive index $n$ both originate from the same complex dielectric response function $\epsilon^{1/2} = n + i\kappa$, where the imaginary part is related to the absorption coefficient by $\alpha = \frac{4\pi E}{hc}\kappa$. As such, absorptance and reflectance are related by the Kramers–Krönig causality relation.[52] A photo-induced change in the absorptance may, thus, be accompanied by a change in the reflectance,[53,54] which would disqualify the (often implicit) assumption of constant reflectance.

A few studies performed simultaneous measurements of time-resolved reflectance and transmittance spectroscopy in order to demonstrate the impact of reflectance change $\Delta R$ on the estimation of the absorptance change $\Delta A$.[32,55–57] Two key results must be highlighted with respect to temperature measurement, as properly taking into account $\Delta R$:

(1) leads to a broadened $\Delta A$ spectrum (as compared to the $\Delta A = -\Delta T$ approximation). This means that the apparent temperature of the spectrum, as measured with an exponential fit, increases due to reflectance change;
(2) reduces or even cancels the high-energy PIA that is sometimes attributed to bandgap renormalization (see Sec. II D 3).

In light of these observations, Price *et al.*[32] measured simultaneously $\Delta R$ and $\Delta T$, showing a significant contribution of the reflectance change [Fig. 4(a)].

Neglecting this contribution (i.e., assuming that the change in transmission results only from a change in absorptance) leads to an underestimation of the temperature by 100–250 K, making it drop below ambient [Fig. 4(b)].

In the literature of hot carriers in perovskite absorbers, this underestimation appears to be often partially compensated by the overestimation resulting from the invalid assumption on effective masses [see Eqs. (8) and (9)].

#### 5. Change in the absorption coefficient and many-body effects

In some experimental conditions, many-body effects cannot be neglected, and a more precise description of the absorption coefficient becomes necessary to account for the TAS signal. For instance, at elevated fluences, the bandgap of a material will be reduced due to bandgap renormalization.[17] In such a context, the variations of the absorption coefficient are more subtle than in Sec. II C,

$$\Delta\alpha(E) = \alpha(E, \Delta E_g)[1 - f_h(E_h) - f_e(E_e)] - \alpha_0(E), \tag{11}$$

where $\Delta E_g$ is the change in bandgap due to bandgap renormalization.

A similar formula was used by Shah in his seminal study of continuous absorption of GaAs in the presence of optical pumping.[35] In the context of TAS, such a form was first used in the context of graphite.[58] The authors used the following approximation: $\alpha(E, \Delta E_g) = \alpha_0(E + \Delta E_g)$. Then, by using tabulated values of the absorption coefficient, they could fit their TAS spectra across an energy window of more than 700 meV. In addition, they showed that a standard fit with high-energy Boltzmann distribution on the same dataset would lead to massive overestimation of carrier temperature, up to a factor 3.

In some experimental contexts, it may be necessary to take into account other many-body effects. For instance, (i) exciton screening is common in perovskite and low-dimensional samples[20,49,59] while (ii) linewidth broadening due to many-body effects can become apparent at high fluences in a wide range of semiconductors.[49,60]

A comprehensive model with linewidth broadening, exciton screening, and bandgap renormalization was used to extract carrier temperatures in bulk and nanoparticle perovskite films.[49] Interestingly, a regular Boltzmann fit was also applied to the data and led to an overestimation of the temperature by about 100 K, with only little change of the overall temperature dynamics. In contrast, the full fit developed in Ref. 32 allowed us to resolve a longer time dynamics that was invisible with the classic mono-exponential fit. Overall, the full fit approach is preferable for at least three different reasons: (i) it is not sensitive to the choice of fitting interval, (ii) it is necessary to reliably measure electron and hole chemical potentials in case of bandgap renormalization, and (iii) it provides deeper understanding of the physical processes at play during carrier relaxation. This approach also has drawbacks: (i) building an accurate physical model for the absorptivity can be challenging in complex structures (stacks) or even impossible (with inhomogeneous quantum dots for example), (ii) the large number of fitting parameters required raises the risk of overfitting.

Pump-probe absorption spectroscopy is a technique that allows us to measure the carrier temperature quite straightforwardly. It has been extensively used in time-resolved regime, but can also be used in the steady-state regime in principle.

With the advent of ultrafast lasers, transient absorption spectroscopy has become a routine characterization technique. However, many assumptions need to be carefully ensured before measuring carrier from transient absorption spectroscopy. These approximations are summarized in Fig. 5 under the form of a decision tree. First, reflectance changes need to be checked and taken into account if non-negligible. Second, the monoexponential approximation can only be rigorously applied when electrons are much lighter than holes and when absorptance is sufficiently small. In all other cases, *ad hoc* full fit models are highly recommended, as they can account for complex optical models as well as many-body effects, which are common in high injection regimes.

## III. PHOTOLUMINESCENCE SPECTROSCOPY

### A. Working principle

Photoluminescence (PL) experiments involve exciting a material with light so that electrons are promoted to higher energy states and then recording the light emitted as these excited states

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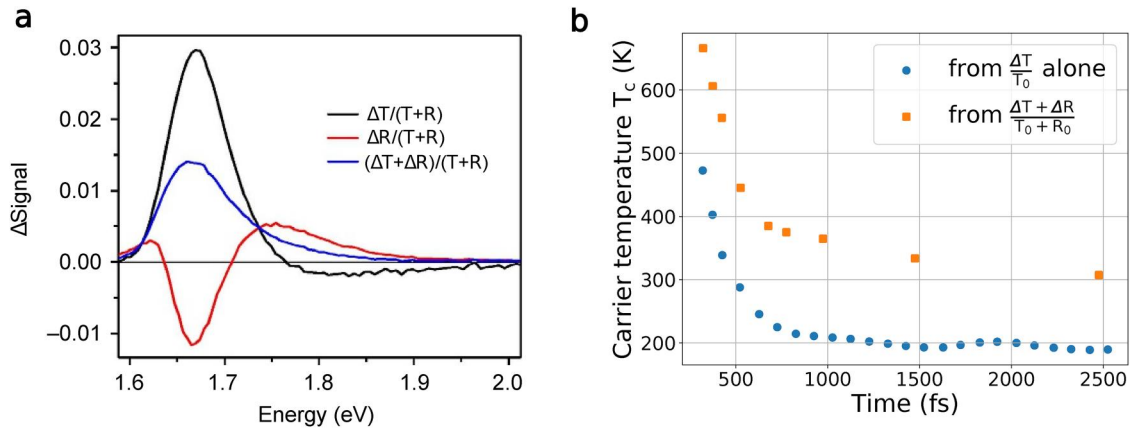



**FIG. 4.** (a) Comparison of transmittance and reflectance changes (and their normalized sum) 5 ps after the pump. Adapted from Ref. 32 [Price *et al.*, Nat. Commun. **6**, 8420 (2015). Copyright 2015 Authors, licensed under a Creative Commons Attribution (CC BY) license]. (b) Carrier temperature decay obtained from transmittance change alone [blue, reanalyzed from Ref. 32 with Eq. (9)] and from both transmittance and reflectance changes (orange).

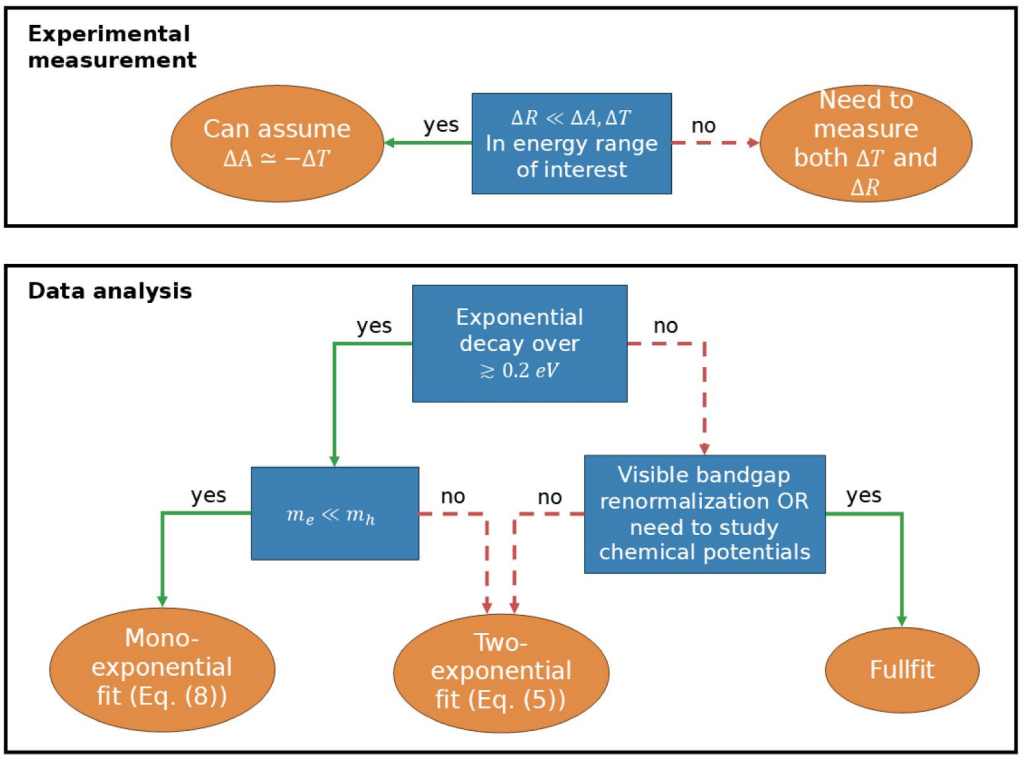



**FIG. 5.** Decision tree to decide which experimental and fitting techniques should be used to extract carrier temperature in the context of TAS. Details can be found in Secs. II C and II D.

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relax back toward equilibrium. The spectral analysis of this emitted light can be leveraged to measure lattice temperature, carrier temperature, or both. This section presents the basic physical understanding necessary to extract carrier and lattice temperatures from PL spectra.

The law of emission of a homogeneous slab of semiconductor, generally referred to as the "Generalized Planck Law" (GPL), has long been derived in the case where electrons and holes are at thermal equilibrium,[61]

$$\phi_{PL}(E) = \frac{2E^2}{h^3c^2} A(E) \frac{1}{\exp\left(\frac{E-\Delta\mu}{k_B T_c}\right) - 1}, \tag{12}$$

where $\phi_{PL}$ is the number of photons emitted per unit of time, surface, energy and solid angle, $E$ is the energy of the emitted photon, $\Delta\mu$ is the quasi-Fermi level splitting (QFLS) and $T_c$ is carrier temperature. We make use of the external absorptance of the system, $A$, which includes the contribution from the reflectance of the sample. $h$, $c$, and $k_B$ are, respectively, Planck's constant, the speed of light, and Boltzmann's constant.

This emission spectrum follows a blackbody baseline governed by the carrier distribution parameters ($T_c$, $\Delta\mu$), which is modulated by the absorptance profile. Because it measures photons emitted by electron–hole recombination, PL is a technique that fundamentally measures the distribution of carriers within their energy bands, which makes the determination of carrier temperature straightforward in principle [see the explicit dependency in $T_c$ in Eq. (12)]. In the case of lattice temperature, the dependency is more implicit and can be carried out thanks to the dependency of the absorptance term on the lattice temperature. This explains that techniques to determine carrier temperature are more developed than those for lattice temperature, as will be discussed in Secs. III C and III D.

Another remarkable feature of the GPL is the presence of the QFLS, $\Delta\mu$, in the denominator, which behaves as a positive chemical potential for the photon distribution. This term appears as an exponential amplification of the PL signal (as long as $\Delta\mu \ll E_g$).

## B. Experimental setup

In this section, we will first present a typical setup for temperature analysis using continuous-wave PL (CWPL) measurement. Then, we will present its time-resolved counterparts. In both cases, it is necessary to use a spectrometer to analyze the spectral shape of the PL signal.

### 1. CWPL spectroscopy

CWPL or steady-state PL spectroscopy relies on the use of a continuous laser to excite a sample. The light emitted by this sample is separated from the laser signal by an optical filter (usually longpass, but sometimes notch or even shortpass) and then guided toward a spectrometer. An example of such a setup is provided in Fig. 6.

Because of the above-mentioned spectral separation, it is not possible to observe the PL spectra close to the laser energy. Therefore, when trying to measure carrier temperature, one needs to ensure that the laser energy is sufficiently high (typically several hundred meV) above the bandgap of the material to see the high-energy tail without laser pollution. In addition, using such a high-energy laser ensures good absorption of the excitation light and, therefore, strong PL signal.

### 2. TRPL spectroscopy

Time-resolved PL (TRPL) spectroscopy consists in measuring both the temporal and spectral evolution of the PL intensity after a pulsed excitation. To probe hot-carrier dynamics, resolutions of about a picosecond or below are typically required.[27] These can be obtained by two main routes: (i) optical gating or (ii) streak cameras.

For optical gating, the excitation laser is split to obtain both a pump and a gating pulse with a controllable delay between them. On the PL detection path, the light goes through an optical gate acting as a shutter that lets the signal through only when the gating beam impinges on it.[27] This can be achieved by using parametric upconversion, where the PL and gating beams are focused on a non-linear crystal and only the sum-frequency signal is collected. This situation is represented in Fig. 7(a). Alternatively, optical gating can be performed by using the Kerr effect.[62] By definition, the temporal resolution of such optical gating systems is given by the width of the laser pulse. Therefore, temporal resolutions of 100 fs are quite common.[63]

On the other hand, streak cameras allow temporal detection thanks to a time-dependent spatial deviation of light, similar to the working principle of oscilloscopes. Typically, an optoelectronic streak tube deviates photoelectrons ripped off from a photocathode with a sweep voltage. The deviated electrons are converted back to photons by a phosphor screen before being imaged on a 2D photodetector. The captured signal is analyzed as a function of time and one dimension in space. When coupled to a spectrometer, as pictured in Fig. 7(b), the spatial dimension directly conveys the spectral information. Modern streak cameras can access sub-picosecond temporal resolutions.[64]

As mentioned before, both CWPL and TRPL spectra can be analyzed with the same strategies. Several methods are known and used in the literature to extract temperatures from a PL spectrum.[22,65,66] In the ideal case, simplifying assumptions can be used to allow for simple determination of the carrier and/or lattice temperature, as will be discussed in Sec. III C. Advanced methods allowing for more accurate temperature determination will be presented in Sec. III D.

## C. Data analysis in the ideal case

In this section, we assume that one or several PL spectra were acquired carefully, and present the most basic techniques to extract carriers and lattice temperature from it. For guidelines regarding data acquisition, or advanced fitting techniques, refer to Sec. III D.

### 1. Carrier temperature

To simplify the analysis of carrier temperature, it is often assumed that the absorptance $A$ in the GPL [Eq. (12)] saturates: $A(E) = A_0$ for $E$ sufficiently above the bandgap. This assumption forms the basis of the historical method to determine carrier

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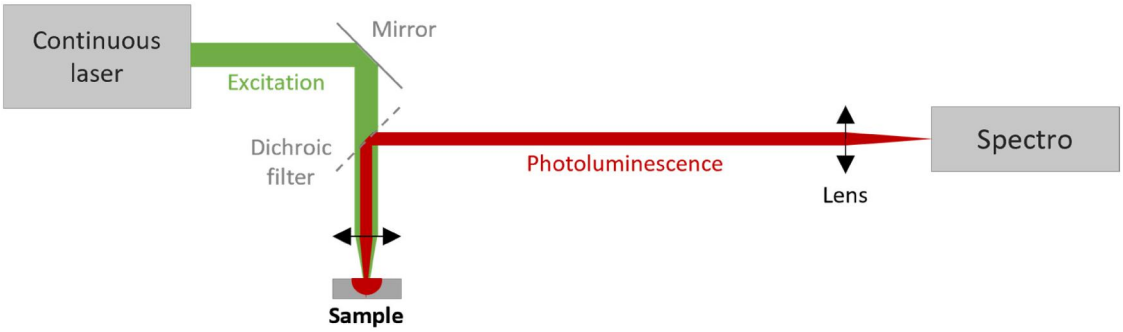



**FIG. 6.** Typical CWPL spectroscopy setup containing a continuous laser, a longpass dichroic filter to separate the low-energy PL signal from the high-energy laser, and a spectrometer to analyze the spectral shape of the emitted light.

temperature from PL measurement, the so-called "high-energy linear fit" (HELF). It was first proposed by Shah *et al.*[67] to study hot-carrier effect in a GaAs sample at cryogenic temperatures.

Under this assumption, the PL spectrum decreases exponentially at high energy $E - \Delta\mu \gg k_B T_c$), with a typical energy scale controlled by carrier temperature $T_c$. Indeed, the GPL can be rewritten as

$$\ln\left(\frac{h^3 c^2 \phi_{PL}(E)}{E^2}\right) = -\frac{E}{k_B T_c} + \frac{\Delta\mu}{k_B T_c} + \ln(A_0). \tag{13}$$

This equation can be fitted by ordinary least square to obtain the carrier temperature. HELF is, by far, the most used method in literature to determine carrier temperature, both for CWPL[67–71] and TRPL measurements.[72–75] A graphical representation of this fitting technique is given on PL data measured on InGaAsP quantum well (QW) in Fig. 9(a).

The assumption of saturated absorptance is usually valid in standard bulk semiconductors (Si, GaAs, etc.), which are optically thick. However, it may fail in weakly absorptive materials such as ultrathin absorbers or quantum structures.

Note that to use this technique, it is necessary to perform at least a relative calibration (also called spectral calibration) to measure the *shape* of the PL spectrum accurately. Absolute calibration, which aims at determining the absolute intensity of the PL emission (in $\mathrm{ph\,s\,m^{-2}\,sr^{-1}\,eV^{-1}}$ for instance), can also prove useful, for instance, to determine the QFLS of the system[76] or to go beyond HELF (see Secs. III D 3–III D 5).

### 2. Lattice temperature

Lattice temperature can be assessed from PL measurements because the absorptance of a material depends on its bandgap,

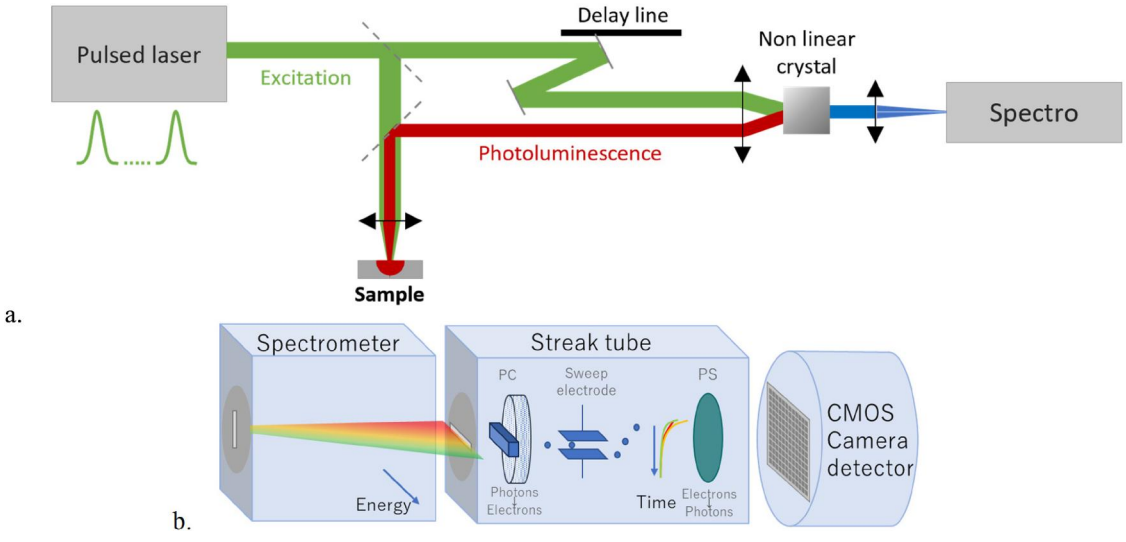



**FIG. 7.** Principle of time-resolved photoluminescence spectroscopy—(a) Example of optical gating based on up-conversion. (b) Scheme of principle of a streak camera. PC (respectively, PS) stand for "photocathode" (respectively, "phosphor screen").

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which itself depends on the lattice temperature. The quantitative link between the absorptance and lattice temperature depends on the geometry of the sample and can be quite complex. However, as a proof of principle, Eq. (3) can be used to describe the absorptance of a homogeneous slab of direct bandgap semiconductor as a function of its absorption coefficient and thickness. The absorption coefficient $\alpha$ itself depends on the bandgap of the material,[77] which in turn depends on lattice temperature $T_L$, typically following the so-called Varshni law:[25]

$$E_g = E_g(T = 0K) - \frac{\alpha T_L^2}{\beta + T_L}, \tag{14}$$

where $\alpha$ and $\beta$ are two phenomenological coefficients.

In practice, regardless of whether these ideal relations hold, it is possible to tabulate the variation of the PL spectrum with lattice temperature. To do so, the sample is placed in a cryostat, and the PL peak energy $E_{peak}^{PL}$ is measured a function of the imposed lattice temperature $T_L$. This measurement must be conducted with a weak enough laser excitation so that the laser does not unintentionally heat the sample or the carriers. Thanks to this preliminary calibration, the observation of the PL peak energy can be used as a proxy of the lattice temperature. The precision of this method depends on the temperature sensitivity of the absorber's bandgap, which is typically in the range of 0.5 meV/K.[78] For such values, a precision 1–5 K has been reported.[66]

## D. Limits, biases, and possible corrections

Despite the apparent simplicity of the HELF method, measuring a temperature that makes physical sense from PL spectra often requires careful examination. Indeed, HELF suffers from many limitations that we will examine in this section. For each limitation, we discuss possible correction through a more advanced fitting of the PL spectrum. The various techniques described in this section are summarized graphically and applied to the same PL spectrum in Fig. 9. A summary of this section under the form of a decision tree, to help choosing which technique to use, is given in Fig. 10.

### 1. Data acquisition

Regardless of the fitting method, the estimation of carriers' temperature is very sensitive to the shape of the PL spectrum. Therefore, it is of prime importance to conduct the data acquisition as cleanly as possible. In practice, this means:

- Ensuring a high enough signal-to-noise ratio. To meet the hypothesis of HELF, it is often necessary to perform the fitting far above the bandgap, in a region where PL signal decreases exponentially. To enhance the signal-to-noise ratio, it may be necessary to resort to "high dynamical range" techniques, where different spectra acquired at different exposure times are stitched together.[79]
- Performing background subtraction carefully. Most photodetectors are subject to background noise. When incorrectly subtracted, background will affect the shape of the PL spectrum in regions of low PL emission, hence also affecting the estimation of carrier temperature.
- Calibration (relative or absolute) of the PL setup can be a great help to avoid distortions of the spectrum due to the spectral response function of the collection system. This is particularly critical for HELF and full fit techniques (see below).

### 2. Fitting interval

Although HELF is a simple method, one must pay attention to the quality of the signal and to the fitting interval that was used. Indeed, to meet the assumption of saturated absorptance, the fitting interval must be located far above the bandgap. However, in such a region, the signal-to-noise ratio of the PL can become weak. In addition, parasitic PL emission from unwanted compounds or other layers in the material may pollute the analysis at high energy. As a sanity check for the validity of HELF, it is advised to estimate how sensitive is the temperature determination to the choice of the fitting interval. If no interval of at least 100–200 meV can be found to perform the HELF, more advanced methods may be necessary.[80]

### 3. Non-saturated absorption

Often enough, the absorptance will not be saturated even at high energy. This is notably the case of ultrathin or quantum structures (quantum wells, wires or dots) with low absorptance and several absorption steps. In this case, HELF does not apply and could lead to errors from about 50 K[23] to several hundreds of kelvin.[65,80] However, it may still be possible to extract a carrier temperature, provided that the absorptance $A$ does not depend on the excitation condition (e.g., excitation intensity in CW or time delay in TR).

On the one hand, it may be possible to estimate the absorptance directly, from external quantum efficiency (EQE) measurements for instance. In this case, an equation like Eq. (13) can be used to fit the carrier temperature at high energy, using the full shape of the absorptance.

On the other hand, even if the absorptance is unknown, the so-called "PL ratio" (PLR) method may be used. This method requires a collection of $N$ PL spectra $\phi_{PL}^{(1)}$, $\phi_{PL}^{(2)}$, …, $\phi_{PL}^{(N)}$ acquired by varying one experimental condition (for instance excitation intensity), and for which the absorptance is identical over a certain energy window. If $A^{(1)} = A^{(i)}$, then for $E \gg \Delta\mu^{(i)}$,

$$\ln\left(\frac{\phi_{PL}^{(i)}}{\phi_{PL}^{(1)}}\right) = -\frac{E}{k_B} \times \left(\frac{1}{T_c^{(i)}} - \frac{1}{T_c^{(1)}}\right) + \frac{\Delta\mu^{(i)}}{k_B T_c^{(i)}} - \frac{\Delta\mu^{(1)}}{k_B T_c^{(1)}}. \tag{15}$$

In this case, the PL ratio has an exponential behavior, whose slope is controlled by the difference between the carrier temperature $T_c^{(i)}$ and the reference carrier temperature $T_c^{(1)}$. By choosing carefully the "reference" spectrum $\phi_{PL}^{(1)}$, we can ensure that $T_c^{(1)}$ is known—for instance equal to the ambient temperature if we select a spectrum acquired with low enough excitation intensity. From here, we can deduce the temperature of all spectra. A graphical example of this technique is provided in Fig. 9(b).

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PLR has been used mostly to study CWPL spectra.[81–83] In principle, it could be applied to a collection of TRPL spectra acquired at different delay times as well. However, TRPL data usually show strong band filling contribution at short time delays, which disrupts the assumption of invariant absorptance (see Sec. III D 4 for details).

### 4. Band filling

Band filling is a common effect in hot-carrier experiments, to the point where some techniques to probe hot-carrier temperatures rely exclusively on the analysis of band filling (like TAS, see Sec. II A). In the context of PL analysis however, band filling is an effect that makes the analysis of carrier temperature more difficult. Indeed, band filling makes the absorptance $A$ dependent on carrier energy $E$, and, therefore, HELF would lead to a systematic error. In addition, band filling is a carrier-induced effect, which depends strongly on carrier density (and, hence, on excitation power and/or time delay), and, therefore, PLR cannot be used either.

Band filling can sometimes be misinterpreted as an elevation in carrier temperature, as it affects the slope of the PL spectrum up to several hundred meV above the bandgap. As an illustration, several simulated PL spectra at room temperature $T_L = T_c = 300$ K, with increasing QFLS $\Delta\mu$, are reported in Fig. 8(a). An estimate of the carrier temperature obtained from HELF is reported in Fig. 8(b) as a function of the QFLS. As the QFLS increases, band filling becomes significant and the estimated carrier temperature drifts away from its real value by 5 K when QFLS reaches the value of the bandgap. Note that band filling will become even more predominant at larger carrier temperatures, inducing errors higher than 30 K for $T_c \simeq 500$ K.

However, in the case where band filling is the only carrier-induced effect affecting the PL spectra, and assuming that the absorptance is low ($A \propto \alpha$), it can be shown that[79]

$$\frac{\phi_{PL}^{(i)}}{\phi_{PL}^{(1)}} = \frac{BF\left(E, T_c^{(i)}, \Delta\mu^{(i)}\right)}{BF\left(E, T_c^{(1)}, \Delta\mu^{(1)}\right)} \times \frac{\left[\exp\dfrac{E-\Delta\mu^{(i)}}{k_B T_c^{(i)}} - 1\right]^{-1}}{\left[\exp\dfrac{E-\Delta\mu^{(1)}}{k_B T_c^{(1)}} - 1\right]^{-1}}, \tag{16}$$

where $BF(E, T, \Delta\mu) = 1 - f_e - f_h$ is the band filling term, as defined in Eq. (5).

This equation forms the basis of the "PL ratio with band filling" (PLRBF) technique and can be fitted to extract the temperature $T_c^{(i)}$ if the temperature of the reference spectrum $T_c^{(1)}$ is known. This technique is represented in Fig. 9(c).

It has been used in a modified form in Ref. 22 to compute carrier temperature in an InGaAs multi-quantum well which exhibited strong band filling effect.

### 5. Change in the absorption coefficient

At elevated excitation intensities, materials can be subject to various many-body effects, in addition to band filling.[84] For instance, "linewidth broadening," "bandgap renormalization," and "exciton screening" have already been introduced in Sec. II A.

These three effects affect mostly the sub-bandgap part of the PL spectrum and, therefore, are particularly detrimental to the ability to determine lattice temperature from the PL peak position (see Sec. III C 2). However, they are also detrimental to the ability to measure carrier temperatures because they superimpose over the band filling effect, making its contribution difficult to assess.

Carrier temperature may still be obtained despite these many-body effects by applying a so-called "full fit" method. This term refers to various techniques where the absorptance is modeled in the entire range of PL emission.[59,85] Proper full fit analysis is more versatile than HELF, PLR, or PLRBF because it can accommodate for as many different many-body effects as one needs. An example of full fit is provided in Fig. 9(d).

For instance, Nguyen *et al.* analyzed the luminescence of an InGaAsP quantum well by accounting for band filling, bandgap renormalization, and linewidth broadening.[24] Lim *et al.* used a similar model for perovskite luminescence.[86] Cao *et al.* used a simpler full fit model that does not take band filling into account to study PbS quantum dots.[87] Full fit has also been used to study TRPL data from GaAs quantum wells.[85,88,89] Note that, with a proper full fit model, it may even be possible to distinguish electron from hole temperatures.[19]

The biggest limitation of these full fit approaches is their huge number of fitting parameters (typically 10 or more). This raises a concern of overfitting: it is not always possible to ensure that the fit converges toward physically meaningful parameters. This overfitting problem can be mitigated by two different strategies. First, absolute calibration can be beneficial to obtain reasonable bounds for the QFLS of the system, thereby allowing for a more precise quantification of the band filling magnitude.[22] Second, some of the parameters can be determined from preliminary measurements such as absorptance or EQE.[19,24] Finally, full fit is more robust when applied to a large collection of spectra, as the physical relation between spectra can help us to constrain the parameter space. For instance, if several increasing excitation intensities are measured, we expect that the carrier temperature, QFLS, linewidth broadening, and bandgap renormalization must all be increasing functions of the excitation intensity.

### 6. Lattice heating

The determination of lattice temperature from the PL peak energy is sometimes a goal itself.[90] However, when trying to measure carrier temperature, lattice heating is a detrimental effect as it modifies the absorptance $A$. If bandgap renormalization and/or linewidth broadening are also significant, disambiguating the contribution from each phenomenon may prove difficult. The most favorable case is when the PL peak position does not change under the influence of excitation. In this case, lattice temperature likely stays constant, and the estimation of carriers' temperature can be trustworthy.

As a conclusion, PL is a valuable tool to measure both carrier and lattice temperature. Carrier temperature analysis has been more developed in the literature. Several techniques exist to do so, which can be applied to both CWPL and TRPL spectroscopy. These techniques, which correspond to different approximations, are represented in Fig. 9. In this case, HELF appears to provide a

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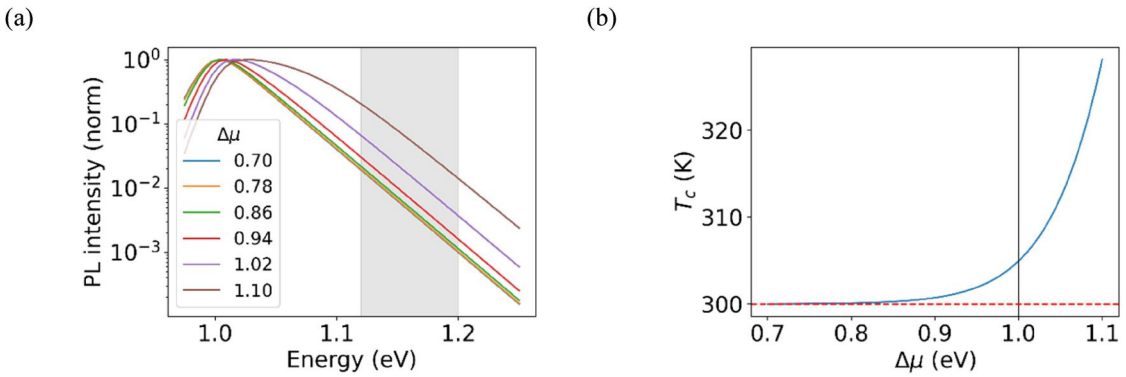



**FIG. 8.** Effect of band filling on PL spectra and on the determination of temperature using HELF method. (a) PL spectra simulated from the GPL [Eq. (5)] for different values of the QFLS, using an ideal step-like absorption coefficient $\alpha_0(E)d = \left[1 + \exp\left(-\frac{E-E_g}{\Gamma}\right)\right]^{-1}$ with $E_g = 1$ eV, and $\Gamma = 8$ meV is a broadening parameter. (b) Carrier temperature as measured using HELF method as a function of the QFLS. The dashed red line indicates the true carrier temperature, and the solid black line indicates the material bandgap.

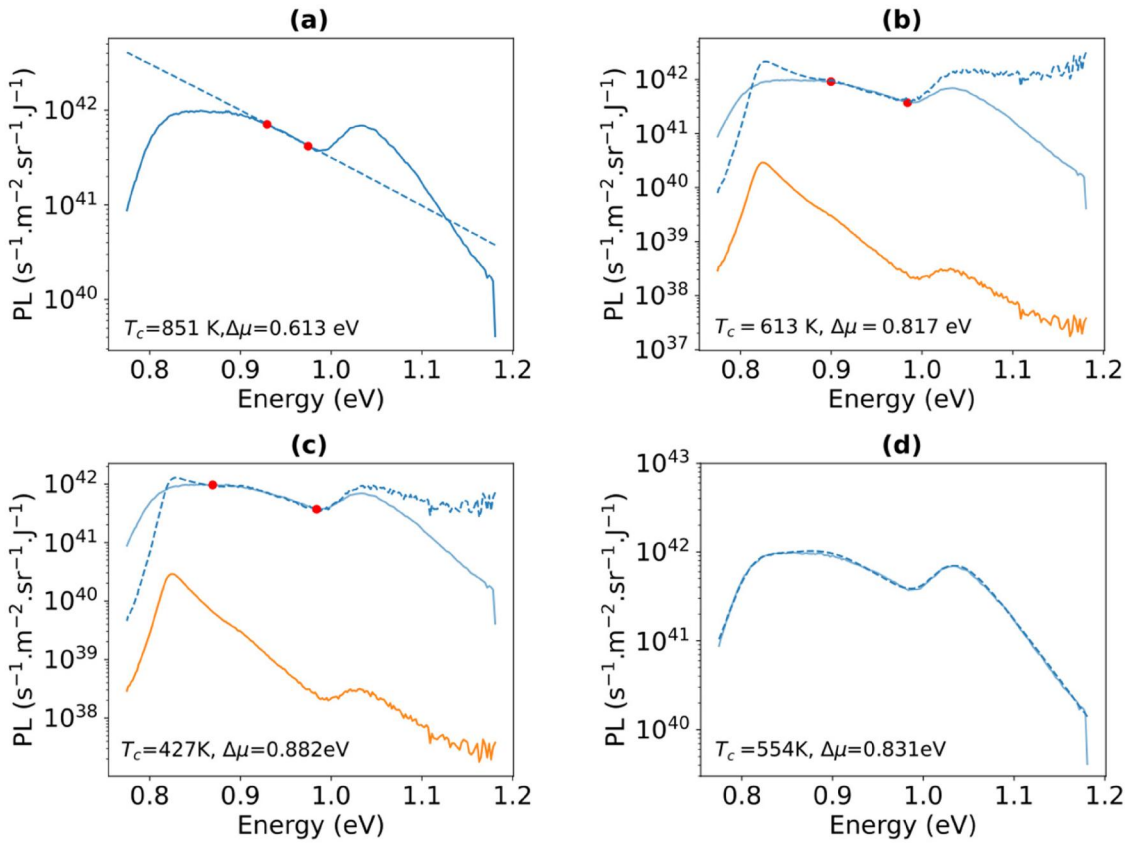



**FIG. 9.** Different fitting techniques applied on the same PL spectrum from an InGaAsP quantum well described in Ref. 19. (a) HELF (assuming $A_0 = 0.01$), (b) PLR, (c) PLRBF, and (d) full fit. Red dots indicate the fit boundaries in each case. Note that for the PLR (respectively, PLRBF) method, the dashed spectrum is obtained by multiplying the fitted model for the PL ratio—given by Eq. (15) [respectively, Eq. (16)]—by the reference spectrum (in orange).

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decent approximation of both the carrier temperature and the QFLS [see Fig. 9(a)]. However, the fitting energy range is only about 0.05 eV due to the emission from the barrier around 1.05 eV, which casts doubt over the applicability of HELF. For the PLR method [see Fig. 9(b)], the estimated QFLS is less than 100 meV below the fitting window, which means that band filling is not negligible in this case. This method can, thus, not be applied reliably. The PLRBF method [Fig. 9(c)] leads to an unrealistic estimation of the QFLS, about 50 meV larger than the bandgap. Finally, a full fit approach provides reasonable estimates of both the carrier temperature and the QFLS for this PL spectrum, but necessitates prior knowledge of the sample's EQE.[19,24]

As the different techniques can yield very different carrier temperatures (450–850 K for this particular spectrum), the assumptions that they rely on must be carefully assessed to choose the most appropriate method. To do so, we present a decision tree in Fig. 10.

## IV. RAMAN SPECTROSCOPY

Raman spectroscopy examines inelastic light scattering: Raman scattering. In this optical process, an incident photon interacts with a material and induces a transition to a different energy state, accompanied by the emission of a photon at a slightly lower (Stokes) or higher (anti-Stokes) frequency. These coherently scattered photons yield information on the material excitations, which could be electronic, vibrational, magnetic excitations, and provide particularly the material structure via its phonon fingerprints, near the center of the Brillouin zone.

Raman spectroscopy enables determining specific mode temperatures in two ways:[91]

- Examining the ratio between the two symmetric signatures called the anti-Stokes and Stokes peaks.
- Monitoring the signature peak shift with temperature increase.

This part will focus on the approach analyzing the ratio of anti-Stokes and Stokes peak intensities to determine the temperature and discuss its uses and limits. The peak shift approach is mentioned at the end of Sec. IV A.

### A. Working principle

Raman spectroscopy consists in measuring the scattered intensity spectrum $I(\omega)$ as a function of the Raman shift $\omega$, defined as $\omega = \omega_L - \omega_S$, with $\omega_L$ being the laser frequency and $\omega_S$ being the scattered frequency. This intensity is related to the fluctuations of the electronic density and is, therefore, connected to the imaginary part of the response function, $\chi''$, which provides information about the energy dissipation in the system after perturbation. The following relation can be derived from the fluctuation–dissipation theorem:

$$I(\omega) \sim (1 + n_B(T))\chi''(\omega), \tag{17}$$

where $n_B(T)$ is the Bose–Einstein distribution of a given material intrinsic excitation, at thermodynamic equilibrium, at temperature $T$, given in Eq. (18). The main family of excitations we will focus on in this work are vibrational excitations. We will, therefore, only refer to phonons and phonon temperature, but the following could also be applied to electronic excitations (discussed in Sec. IV D 5) and magnetic excitations:

$$n_B(T) = \frac{1}{e^{\frac{\hbar\omega}{k_B T}} - 1}. \tag{18}$$

Hence, the peak intensity depends on the given phonon population in the material via the occupation probability of the corresponding phonon state at temperature $T$. For a Stokes/anti-Stokes process, and given that the response function $\chi''$ is an odd function, the scattered intensity is, respectively,

$$I_S = I(\omega) \propto (n_B + 1)\chi''(\omega), \tag{19}$$

$$I_{AS} = I(-\omega) \propto n_B\chi''(\omega). \tag{20}$$

In the case of a photon counting detector, the ratio of the anti-Stokes/Stokes intensities results in Eq. (21), with a cubic prefactor, assuming ideal experimental conditions,[92]

$$\frac{I_{AS}}{I_S} = \frac{(\omega_L + \omega)^3}{(\omega_L - \omega)^3}\frac{n_B}{n_B + 1} = \frac{(\omega_L + \omega)^3}{(\omega_L - \omega)^3}e^{-\frac{\hbar\omega}{k_B T}} \approx e^{-\frac{\hbar\omega}{k_B T}}. \tag{21}$$

The second half of Eq. (21) results from the commonly used approximation $\omega \ll \omega_L$ and will be further discussed in Sec. IV D 2. Nonetheless, by taking the ratio of anti-Stokes and Stokes Raman scattering intensities, one can access the phonon population and retrieve the corresponding phonon temperature. Although this method implementation has limits due to the weak anti-Stokes signal, in particular, at low temperatures, it has been successfully applied in various condensed matter systems.[93,94]

A second Raman thermometry approach exploits the fact that Raman phonon mode linewidth and frequency depend on temperature due to both thermal expansion (in quasi-harmonic interactions) and phonon–phonon anharmonic interactions (at fixed volume). In most cases, the phonon mode shifts to lower frequencies (i.e., softens) and its linewidth increases, with increasing temperature. In semiconductors and insulators, the anharmonicity usually dominates the temperature-dependent peak position and line shape (examples in silicon in Refs. 95 and 96), but in metals and semimetals, electron–phonon coupling can be more significant in the temperature effects.

The peak linewidth can often be tricky to carefully characterize due to multiple effects at play that modify the peak shape (strain, defects, electron–phonon interactions), making the choice of a fitting function subject to discussion (discussed in Sec. IV D 1). On the other hand, the peak position as a function of temperature is usually a clear signature, easy to extract, whatever the fit function. However, it can be small (<1 $cm^{-1}$ for 50 K) or even temperature-independent if $k_B T \ll \hbar\omega_0$ and it requires a fine calibration curve, relating the Raman peak position (i.e., phonon frequency) to the real material temperature, which can be subject to heating from the laser (discussed in Sec. IV D 4) or biased direct temperature measurement using sensors.

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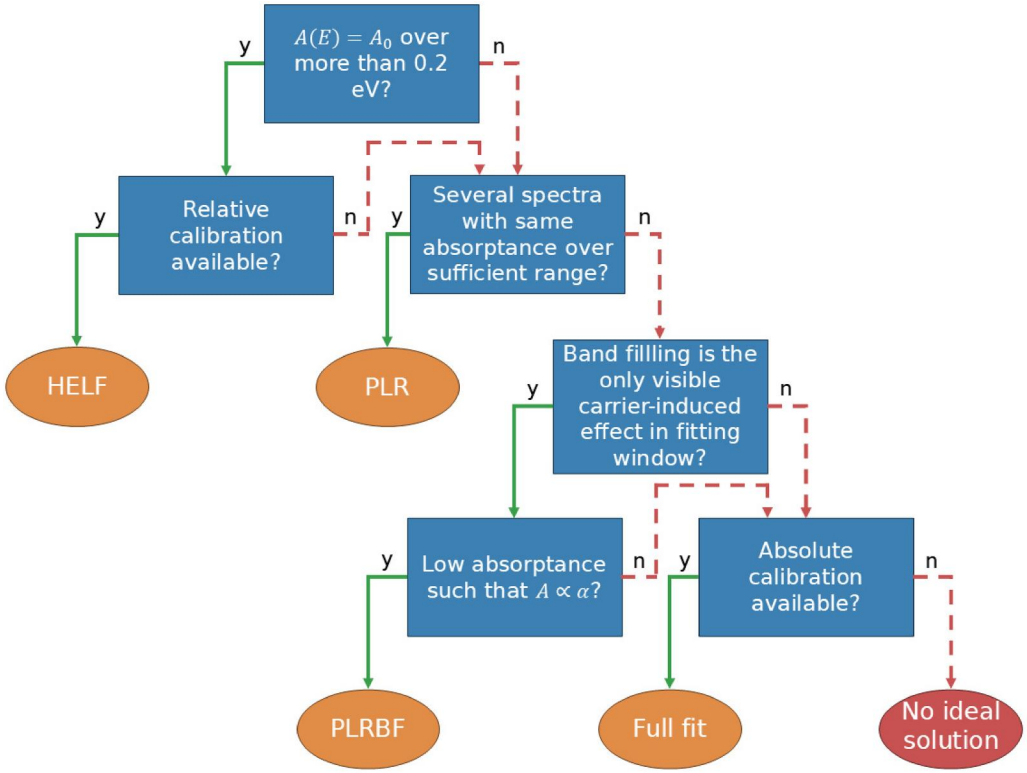



**FIG. 10.** Decision tree to choose which technique to use to extract carrier temperature. Details can be found in Secs. III D 3–III D 5.

Hence, the peak position approach is an easy and precise method of thermometry but is limited to the precision and range of needed preliminary temperature calibration of the peak position, while anti-Stokes/Stokes ratio approach does not rely on such temperature calibration and can, thus, provide high temperature extraction, but is limited at low temperature due to weak anti-Stokes signals.[97] We will detail the anti-Stokes/Stokes ratio approach in the following, as it is more versatile.

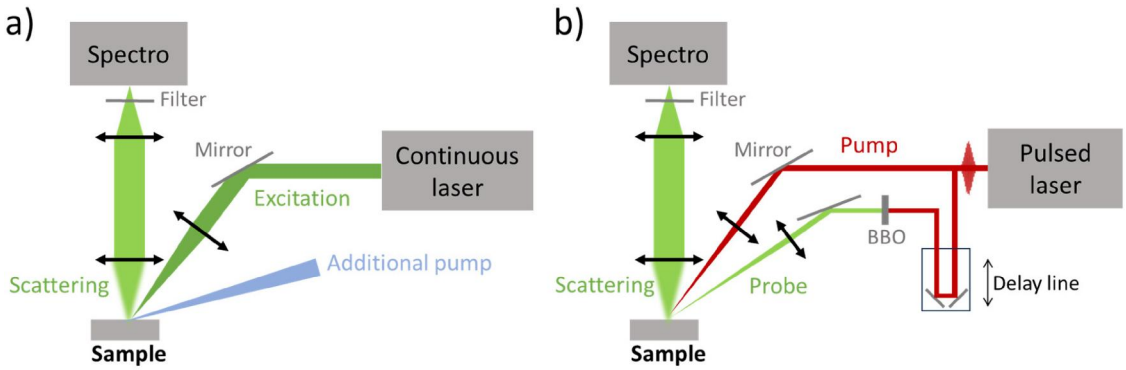



**FIG. 11.** (a) Continuous (CW) Raman spectroscopy typical experimental setup, relying on a narrow CW laser focused on the sample, collection, and refocused to a spectrometer. One can also add a second continuous beam that can act as a pump to study out-of-equilibrium effects (light blue beam). (b) Two-color time-resolved Raman spectroscopy setup, relying on a pulsed laser, separated into two lines, one for pumping the sample, the other one probing the sample at the doubled frequency [after barium borate (BBO) crystal], with a given delay controlled by a delay line. The scattering is collected and focused to a spectrometer.

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We finally note that the ratio anti-Stokes/Stokes as a function of excitation power enables the extraction of the phonon lifetime $\tau_{ph}$, which is a complementary quantity to the phonon temperature,[98] and the absolute phonon population.

## B. Experimental setups

### 1. CW Raman spectroscopy

Continuous Raman spectroscopy relies on the use of a continuous laser, which is focused to excite the sample. The light scattered from the sample is collected by a lens or an objective for larger numerical aperture, guided and focused to a spectrometer. An example of such a setup is provided in Fig. 11(a). Since both Raman and Rayleigh scatterings are collected, it is required to add a filter to cut off the laser frequency, or to rely on a spectrometer with high resolving power, to reject efficiently the elastic Rayleigh scattering. Since the Raman scattering is very weak ($10^{-5}$ * Rayleigh scattering), it is useful to have CCD camera with low dark current and high sensitivity, cooled with liquid nitrogen for example. Half-waveplates and polarizers can be added on the incident and scattered beams to control their polarizations and study the anisotropy of the material response.

As for PL and TAS techniques, the spectral calibration of the spectrometer is essential for the peak identification and accurate peak intensity extraction, accounting for polarization and wavelength dependence of the setup elements (including the grating efficiency and the detector response). This calibration can be achieved with a standard light source (lamp or fluorescent substance), or using a reference signal in the sample under scrutiny.[99] Note that an advantage of Raman spectroscopy is that the spanned spectral range where calibration is required is usually narrower than for PL measurements for example.

Although Raman intensity is very weak, it is possible to measure a relevant signal using reasonable laser power (1–10 mW) and acquisition times (few minutes, depends on phonon signal). To study the material response in its equilibrium state, one can choose a laser energy below the material bandgap, mostly preventing sample heating due to laser absorption, although some absorption channels may remain (defects, Urbach-tail, multiphonon absorption, etc.).

For above bandgap excitation, Raman scattering is enhanced which can be valuable if measuring weak signals. However, parasitic effects also appear, such as sample heating since the laser is absorbed, discussed in Sec. IV D 4, or complexities due the presence of photoexcited carriers which can couple to phonons.[100] Special care must be taken in the case of resonant excitation with a strong interband transition or in the vicinity of the bandgap as Stokes and anti-Stokes responses might not be symmetric anymore.[101] More details are given in Sec. IV D 3.

A second CW beam, acting as a pump, can be added to excite the sample at a different energy and perform out-of-equilibrium Raman spectroscopy, typically to study the material stationary phononic response in the presence of photoexcited carriers.[98] It has also been used by decoupling spatially the two beams to measure the thermal conductivity at different positions from the pump position.[102]

### 2. Time-resolved Raman setup

Time-resolved Raman spectroscopy will give access to Raman spectra at specific delays after driving the sample out of equilibrium by an optical pulse pump. It is of particular interest to highlight individual phonon dynamics, resulting from their coupling to photo-excited electrons and other phonons, which lifetime could typically be on the ps order of magnitude. One possible configuration involves the implementation a two-color setup, with the probe being at the doubled frequency of the pump for example, to easily disentangle the response of the probe. An example of such a setup is provided in Fig. 11(b). The temporal resolution depends on the laser pulse duration. Since a pulse laser is used, the laser bandwidth is much larger than the CW configuration, which leads to a significantly smaller spectral resolution, but this is not an obstacle for extracting anti-Stokes/Stokes ratio and retrieving the phonon temperature. Using tr-Raman spectroscopy is particularly interesting to access each phonon temperature since they can be different in an out-of-equilibrium state, revealing preferential coupling to photo-excited electrons, with ps to sub-ps temporal resolution.[103] One can also estimate the average electron–phonon interaction time from the increase time of phonon population following the photo-excitation of hot carriers, as shown for GaAs in Ref. 104.

Other temporal Raman-based techniques have been developed, such as laser flash Raman[105] or time-domain differential Raman, providing temperature characterization with longer temporal resolution and tunable heating time. These techniques, relevant for probing nanostructure temperatures, are discussed in Ref. 102.

The studies presented below only report CW Raman spectroscopy measurements.

## C. Data analysis and temperature determination

### 1. Data analysis

In Fig. 12, we present the Stokes (right part) and anti-Stokes (left part) Raman spectra, measured on a bulk GaAs sample in a helium cryostat, for two temperatures, at 120 K (blue) and 250 K (orange). A CW laser beam at $\lambda = 532$ nm with a power of 20 mW is focused on the sample to a spot of radius 22 $\mu$m. The spectra show a peak corresponding to a well-known LO phonon mode. This mode red-shifts with increasing temperature, as shown on the spectra, with $\nu_{LO} = 293.9$ cm$^{-1}$ at 120 K and $\nu_{LO} = 291.5$ cm$^{-1}$ at 250 K.

To extract the Stokes and anti-Stokes spectral intensities, we fit the phonon mode peaks with Lorentzian functions, shown with red curves. If the modes present some asymmetry, due to electron–phonon coupling for example, the Lorentzian function might not be the most appropriate choice. This is discussed in Sec. IV D 1.

### 2. Temperature determination

After extracting the intensity of the Lorentzian fits on both Stokes and anti-Stokes spectra, we can retrieve the temperature using Eq. (21). Figure 13 presents the retrieved temperature (orange dots) from analyzed GaAs Raman spectra partially shown

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in Fig. 13, as a function of the sensor temperatures. It is overlapped with sensor temperatures (blue dots). We note an overall good agreement between the retrieved temperature and the sensor temperature. The differences may have several explanations. First, they can come from non-optimal thermal conduction between the cold finger holding the temperature sensor and the sample, where we measure the temperature from Raman thermometry. Second, the probe power, here a fluence of 5 kW/cm$^2$, is absorbed by the sample, potentially resulting in a temperature increase. This second possible bias is discussed in Sec. IV D 4.

## D. Limits, biases, and possible corrections

Once both Stokes and anti-Stokes Raman spectra are acquired using a sensitive spectrometer, several difficulties may arise when extracting the correct sample temperature that can be summarized as follows:

- How to get an accurate anti-Stokes/Stokes ratio? (fitting issues)
- How to get the correct temperature from the ratio?
- What does the extracted temperature really correspond to?

### 1. Difficulties in fitting

For extracting the anti-Stokes/Stokes ratio from Raman spectra, we rely on a fitting procedure of the phonon peak in the spectra, which can be perturbed by other effects:

- Electronic/Photoluminescence background

If the laser energy is above the material bandgap, free carriers are excited leading to photoluminescence (PL), which influences the measured spectrum. To get rid of PL, one can fit the tail of the PL with a decreasing exponential, which is then subtracted to the spectrum to keep only the Raman response (see the "high-energy linear fit" in Sec. II C). In addition to the phonon peaks, electronic Raman responses can be present in the spectra, usually in the form of a background. To have accurate peak intensities, one needs to carefully subtract the electronic background from the spectra.

- Non-Lorentzian fits

In the simplest cases, the shape of phonon peaks should be Lorentzian. However, the presence of anharmonic phonon–phonon interactions and electron–phonon coupling can result in asymmetric peaks. When evaluating the anti-Stokes/Stokes ratio, the peak amplitude will be the important value to extract (provided that the background is correctly subtracted), so the exact phonon line shape should not be critical. However, an inappropriate model (non-physical) or extracting the area under the curve can lead to more errors as discussed in Ref. 94.

### 2. Accurate experimental model and temperature sensitivity

Once the anti-Stokes/Stokes ratio has been properly obtained, we stated before that the temperature can be extracted using Eq. (21), which presents both the accurate expression in the case of a phonon counting detector and the commonly used approximation. We recall that the laser beam induces a polarization in the material, which radiates a power $P(\omega) \propto \omega^4 |p_\omega|^2$, leading to a Stokes intensity $I_S \propto (1+n_B)\omega_S^4$ and an anti-Stokes intensity $I_{AS} \propto n_B \omega_{AS}^4$, with $\omega_{S/AS} = \omega_L \pm \omega$, $\omega_L$ the laser frequency, and $\omega$ the vibrational mode frequency.

With $\omega \ll \omega_L$, we have $\omega_S \approx \omega_{AS}$ and we retrieve the previous approximation shown in Eq. (21). Without this approximation, we obtain the accurate expression for the anti-Stokes/Stokes ratio for detection based on average energy measurement, in ideal conditions,

$$\frac{P_{AS}}{P_S} = \frac{(\omega_L+\omega)^4}{(\omega_L-\omega)^4} e^{-\frac{\hbar\omega}{k_B T}}. \tag{22}$$

For a detection based on photon counting (typically with a CCD camera), the radiated power must be divided by the energy

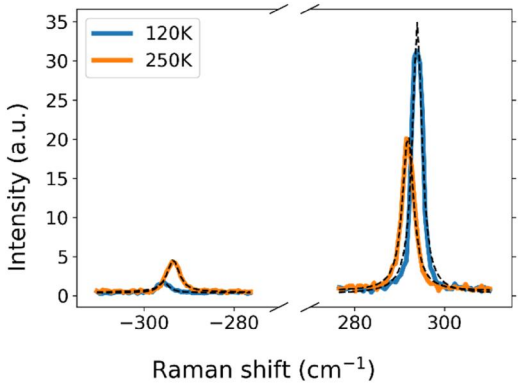



**FIG. 12.** Stokes (right) and anti-Stokes (left) Raman spectra measured on bulk semiconductor GaAs for 120 K (blue) and 250 K (orange). The data are overlapped with Lorentzian fits for each peak (black dashed curves).

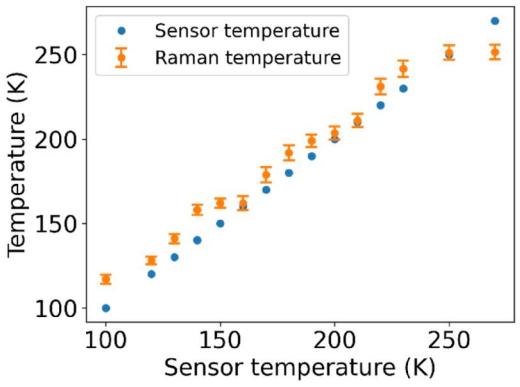



**FIG. 13.** Retrieved GaAs sample temperature from Raman thermometry (orange dots) as a function of sensor temperature, compared with sensor temperature (blue dots); the sample was in a helium cryostat.

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of a photon $\hbar\omega$ to obtain the detected intensity, leading to the expression stated in Eq. (21) with a cubic power.

From a given measured anti-Stokes/Stokes ratio, the three equations will result in different temperatures. This is exemplified in Fig. 14, which presents the temperature dependence of the calculated anti-Stokes/Stokes ratio using the approximate or exact expression in Eq. (21) (blue and orange curve) and the power expression (22) (green curve), for a laser at $\lambda = 532$ nm and a phonon mode at a frequency shift of $\omega = 1000$ cm$^{-1}$. For instance, from a measured ratio of 0.15, T = 615 K could be extracted for an average energy detector, T = 649 K for a photon counting detector, and T = 752 K with the approximative expression, showing how critical it is to use the accurate model, depending on phonon frequency.

Additionally, the choice of phonon mode to perform Raman thermometry has an impact on the temperature sensitivity and, thus, on the extracted temperature uncertainty. What matters here to accurately determine the temperature will be: (1) a good signal-to-noise ratio for the Stokes and anti-Stokes modes, (2) a large enough anti-Stokes/Stokes ratio, and (3) significant variations of the anti-Stokes/Stokes ratio with temperature.

For the first point, we underline that high-energy phonon modes have lower excited state population, resulting in the lower signal-to-noise ratio for the anti-Stokes peak, so choosing a lower energy phonon could be more appropriate for Raman thermometry. Moreover, for high-energy modes, the anti-Stokes/Stokes ratio is very low for temperatures below 300 K (second point), as shown in Fig. 14, but seems appropriate to extract temperature above 300 K.

For lower temperature investigations, lower energy phonons are more suitable as they present a much larger anti-Stokes/Stokes ratio, typically up to 0.98 (not shown here) for a 10 cm$^{-1}$ phonon mode in the 0–800 K range. However, the ratio variations are very small from 200 to 800 K (third point), changing from 0.93 to 0.98, meaning that the temperature sensitivity is very weak in this temperature range and this choice of phonon mode is inappropriate for accurate temperature extraction in the >200 K range.

Overall, if the material under study presents several phonons in different energy ranges, it is convenient to choose the appropriate phonon that corresponds to the temperature range to investigate, to access the lattice temperature, assuming thermal equilibrium between phonons.

### 3. Raman scattering under resonant excitation

As stated before, relations for an anti-Stokes/Stokes ratio as a function of frequency and temperature, given in Eqs. (21) and (22), respectively, for photon counting and average energy detector, correspond to an ideal case, assuming no additional effect (interferences, absorption, polarization or wavelength dependence, etc.).

For excitation energy above the bandgap, transitions happen toward real states in a conduction band continuum (rather than toward virtual states). The laser-induced transition and the scattered transition are resonant with specific energies, leading to an enhancement of the Raman scattering. Resulting sample heating is discussed in Sec. IV D 4. If the excitation happens within a continuum, typically somewhere in the conduction band, with a significant number of states below and above with similar densities of states, Stokes, and anti-Stokes responses should be similar (same resonant prefactors).

However, if the laser energy is resonant with a strong narrow interband transition or with the vicinity of the bandgap, the laser-induced and scattered transitions, involved in either Stokes or anti-Stokes process, might not all be resonant with the given specific material transition. Additional care should be taken to describe separately Stokes and anti-Stokes responses, with respective prefactors containing each possibly resonant denominator, as described in Ref. 101.

### 4. Heating induced by the probe

As previously mentioned, if the probe energy is larger than the bandgap, the probe power is absorbed by the sample and

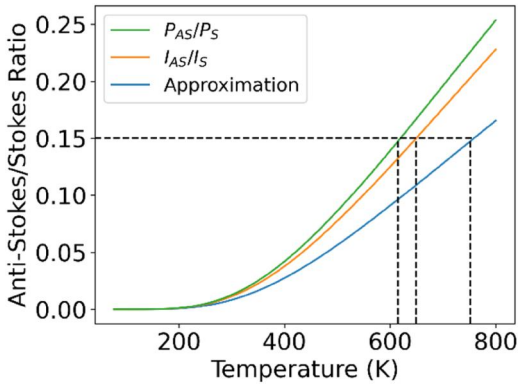



**FIG. 14.** Anti-Stokes/Stokes ratio as a function of temperature, calculated for an average energy detector (green), a photon counting detector (orange) and in the approximation $\omega \ll \omega_L$ (blue), for a phonon mode at $\omega = 1000$ cm$^{-1}$ and a laser at $\lambda = 532$ nm.

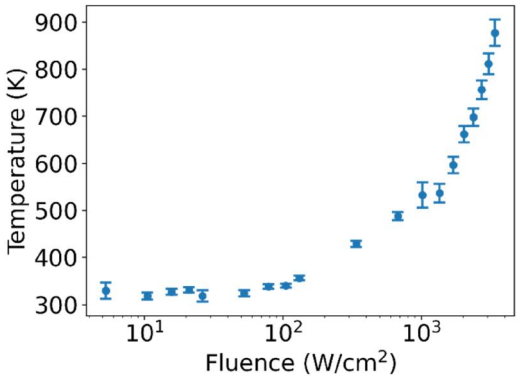



**FIG. 15.** Retrieved GaAs sample temperature from Raman thermometry as a function of probe fluence; the sample was in a room temperature environment.

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**TABLE I.** Comparison between the three thermometry methods discussed in this work.

| Criteria | Photoluminescence (PL) | Absorption spectroscopy (dA/A) | Raman spectroscopy |
|---|---|---|---|
| Probed subsystem | Carriers' temperature | Carriers' temperature | Phonons' temperature |
| Sensitivity | Very high (single-photon detection possible) | Moderate (requires high carrier densities) | Moderate (depends on Raman cross section) |
| Dynamic range | High: $10^{-6}$–1(relative to highest intensity) | Low: $10^{-3}$ to $10^{-2}$ (relative absorption change) | Moderate: $10^{-4}$–$10^{-1}$ (Raman peak shifts/intensities) |
| Key strengths | • Spectral shape reflects carrier temperature in an interpretable way<br>• Works at low carrier densities | • Direct probe of carrier distributions<br>• Can distinguish band filling vs many-body effects<br>• Sub-ps time resolution | • Direct probe of individual phonons and the lattice<br>• Possible symmetry filtering with polarization |
| | Fast acquisition possible in favorable cases (ms), contactless, possible in CW and TR, spatial resolution close to diffraction limit, temperature range from 10 K to over 1000 K | | |
| Key limitations | • Slow intensity decay (sub ns) but fast spectral evolution<br>• Requires large photoluminescence quantum yield (PLQY) to achieve good signal to noise ratio<br>• Advanced fitting may require a good knowledge of absorptance profile (or properties)<br>• Requires a large spectral range for high temperatures | • Sensitive to reflectance changes (often neglected)<br>• Requires transparent, thin samples, leading to heat management issues.<br>• May require assumptions on effective masses<br>• Requires a large spectral range for high temperatures | • Weak Raman signals in some materials<br>• Decreasing accuracy for temperatures much larger or much lower than the phonon energy. |

potentially results in a temperature increase. Figure 15 presents the extracted temperature from Raman thermometry as a function of probe fluence (in W/cm$^2$), for Raman measurements of bulk GaAs (same as before), using $\lambda = 532$ nm probe wavelength. The power and spot size were varied to explore a large range of fluences.

Thus, the effective temperature of the sample can be different from the temperature given by the sensors, emphasizing the importance of estimating the real temperature of the material using, for example, Raman thermometry. Moreover, intense probe illumination may rapidly result in photo-degradation as for hybrid perovskites and directly affects the reliability of repeated anti-Stokes/Stokes measurements.

### 5. Extracting the electronic temperature

In the presence of free carriers, Raman spectroscopy also includes electronic scattering, which often appears as a continuum in the spectrum. One can, in principle, also use the same principle of anti-Stokes/Stokes ratio to extract a temperature from an electronic signature, if both anti-Stokes and Stokes signals are present. The difficulty lays in the origin of the continuum, which can be due to electronic scattering but also to residual fluorescence, which makes it difficult to extract only the Raman scattering contribution to have valid conditions for the anti-Stokes/Stokes ratio.

In the presence of photo-excited carriers, one can link carrier temperature to some phonon temperature for those which couple efficiently to charge carriers through Fröhlich interaction, typically LO phonons.

## V. CONCLUSION

The exploration of temperature in semiconductor systems driven far from equilibrium reveals a landscape far more complex than the simplistic notion of a single, homogeneous thermal state. As this review has demonstrated, direct optical thermometry methods offer a useful set of techniques to access the thermal dynamics of charge carriers and phonons, as soon as the distributions can be effectively described with a temperature. Luminescence spectroscopy, by comparing emitted radiation to the generalized Planck's law, directly accesses the energy distribution of recombining carriers and, hence, their average temperature. In some cases, electron and hole temperatures can be separately accessed. Absorption spectroscopy, through its sensitivity to band filling, provides insights into carrier populations and their thermalization dynamics. Raman spectroscopy, meanwhile, offers a fingerprint of individual zone center phonon distributions and of carrier–phonon coupling.

As shown, each of these techniques has specific advantages and drawbacks, which are summarized in Table I. As optical methods, they all allow a contactless temperature assessment, compatible with operando investigation of materials and devices, such as high-field transistors, LEDs, hot-carrier solar cells, or thermionic coolers. Each method opens a view on a portion of the system, whether occupation of electronic states (TAS) in strongly excited systems, average electron and holes population and temperatures as well as average lattice temperature in TR and CW (PL), or specific lattice excitation (zone center phonons) for

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Raman. Each method has also specific biases and operates under specific assumptions, making their integrated use preferable for a comprehensive understanding of non-equilibrium phenomena in semiconductors.

However, as this review has highlighted, no single technique is universally reliable. The simplicity of exponential fits in transient absorption spectroscopy (TAS), for example, is sensitive to fitting intervals, many-body effects, and changes in refractive index. All these methods may also present issues of sample heating as samples and substrates should be thin while the optical power used is usually large, factors that have to be taken into account when designing the experiments.

Neglecting these factors—such as the contribution of photo-induced absorption or the impact of reflectance variations—or using an unfit model (e.g., overlooking how incoming photon energy is split between electrons and holes), can lead to systematic overestimations of carrier temperatures by factors of 2 or more. Similarly, the assumption of thermalized distributions, while often valid, breaks down in regimes where non-thermal populations dominate. These limitations underscore the need for rigorous cross-validation between techniques, as well as the development of more elaborate models (full fit) that account for multiple physical effects simultaneously.

It is, therefore, essential to identify (i) which subsystem is probed by each technique (carriers, phonons, or lattice), (ii) what are the dominant sources of error (e.g., bandgap renormalization, exciton screening, or reflectance changes) and how to mitigate them, and (iii) check the validity ranges of common data analysis procedures, many of which rely on implicit assumptions that may not hold under all experimental conditions.

In terms of practical recommendations emerging from this review, one should associate several techniques whenever possible, compare simple models with more complete ones (such as full fit) that use the full information available, clearly report the method (e.g., fitting intervals) and the assumptions (e.g., effective mass approximations, thermalization timescales, neglected reflectance changes) in publications. This should facilitate comparisons across studies and accelerate the identification of best practices.

The out-of-equilibrium physics of photo-excited electrons is of particular interest, for example, in hot-carrier systems or in perovskites with strong electron–phonon coupling. These studies would benefit from the combined use of the three discussed techniques, providing access on one hand, to the electron temperature with TAS and with TRPL, and on the other hand, to the probed phonon temperature with transient Raman spectroscopy.

Additionally, both time-resolved and CW versions of the techniques are relevant to describe out-of-equilibrium material, either in a transient or in a stationary state, providing a more complete comprehension of the physics at play and an opportunity to better identify dominant cooling mechanisms.

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## AUTHOR DECLARATIONS

### Conflict of Interest

The authors have no conflicts to disclose.

### Author Contributions

N. Roubinowitz, T. Vezin, and M. Legrand contributed equally to this paper.

**N. Roubinowitz**: Conceptualization (supporting); Investigation (lead); Validation (equal); Writing – original draft (lead); Writing – review & editing (equal). **T. Vezin**: Conceptualization (equal); Investigation (lead); Validation (equal); Writing – original draft (lead); Writing – review & editing (equal). **M. Legrand**: Conceptualization (equal); Methodology (equal); Project administration (equal); Supervision (equal); Validation (lead); Writing – original draft (equal); Writing – review & editing (equal). **S. Der**: Conceptualization (supporting); Investigation (equal); Validation (supporting); Writing – original draft (equal); Writing – review & editing (equal). **B. Yu Jeco-Espaldon**: Investigation (supporting); Writing – original draft (supporting); Writing – review & editing (equal). **S. Rani**: Investigation (equal); Methodology (supporting); Writing – original draft (supporting); Writing – review & editing (supporting). **M. Giteau**: Investigation (supporting); Methodology (supporting); Writing – original draft (supporting); Writing – review & editing (supporting). **Y. Okada**: Methodology (supporting); Writing – original draft (supporting); Writing – review & editing (supporting). **J. F. Guillemoles**: Conceptualization (supporting); Investigation (supporting); Methodology (supporting); Validation (supporting); Writing – review & editing (supporting). **S. Houver**: Investigation (equal); Supervision (equal); Writing – original draft (equal); Writing – review & editing (equal). **D. Suchet**: Conceptualization (equal); Investigation (equal); Methodology (equal); Project administration (equal); Supervision (equal); Validation (lead); Writing – original draft (equal); Writing – review & editing (equal).

## DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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